Face

The portion of the teeth next to the cutting edge is known as face.

Land

The relieved back portion of the tooth adjacent to the cutting edge. Its value is approxmately 1.5 mm. It is relieved to avoid interference between the surface being machined and the cutter.

Outside diameter

The diameter of the circle passing through the peripheral cutting edges.

Centre hole

It refers to hole present at the centre of cutter. A keyway is cut inside the hole.

Cutter angles

Relief angle

It is the angle between land of the tooth and the tangent to the outside diameter of the cutter at the cutting edge of the particular tooth. (Approx 7.5^0)

Primary clearance angle

It is the angle between the back of the tooth and the tangent drawn to the outside diameter of the cutter at the cutting edge (Approx 15^0).

Secondary Clearance Angle

It is the angle formed by the secondary clearance surface and the tangent to the periphery of the cutter at the cutting edge.

Rake angle

The angle measured in the diametral plane between the face of the tooth and a radial line passing through the cutting edge of the tooth. The rake angles may be positive, negative or zero. If the face and the tooth body are on the same side of the radial line and the tooth body may be on opposite sides of the radial line. Then the rake angle is negative. If the radial line and the tooth face coincide in the diameter plane the rake angle is zero.

5.14 Milling cutter material

The milling cutters are generally made of the following materials.

1. Tool steel – High Speed Steel (HSS)
   High Carbon Steel (HCS)
2. Cemented carbide
3. Stellite

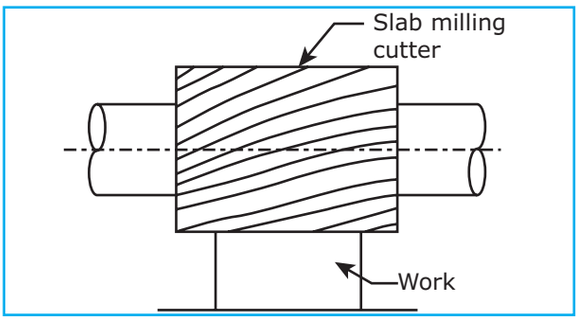
> In general shop work, the high speed steel cutters are most widely used.

5.15 Milling machine operations

The following operations are performed using suitable milling cutters

Horizontal milling

It is the operation of production of a flat surface parallel to the axis of rotation of the cutter. It is also called as slab milling, plain milling cutters and slab milling cutters are used to perform this operation.



Horizontal milling

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